FinTrust Digital Bank - Data Analytics Track

Week 4, Part D: Final Business Insights

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Six final insights from 1,500 customers and 12,000 transactions (January to March 2026). Each follows Finding, Evidence, Business Meaning, Recommended Action. Every figure was recomputed from the raw data for this document, and the key comparisons were checked with significance tests (chi-square and binomial) so that only differences larger than random variation are presented as findings.

Insight 1: International transactions are flagged for risk review about twice as often	
Finding	International transactions have a risk-review rate of 36.9%, against 18.9% for domestic ones.
Evidence	177 of 480 international transactions were flagged (36.9%) versus 2,175 of 11,520 domestic (18.9%). The difference is statistically significant (chi-square $p < 0.001$). International transactions are only 4.0% of volume but 7.5% of all 2,352 flagged transactions.
Business Meaning	Cross-border activity is the clearest risk signal in the data. The review team spends a disproportionate share of effort on a small slice of traffic.
Recommended Action	Apply a dedicated review rule to all international transactions and track the international flag rate monthly as a risk KPI. Note: ATM + International is the highest cell at 46.0% (63 transactions, 95% interval about 34% to 58%), but channel differences within international are not statistically significant ($p = 0.27$), so treat the ATM peak as a lead to monitor, not a proven hotspot.

Insight 2: Risk-review rate climbs steeply with transaction value	
Finding	Larger transactions are flagged much more often: about 18% below NGN 50K, rising to 41% above NGN 250K.
Evidence	Flag rate by value range: 0-10K 17.7%, 10K-50K 17.8%, 50K-100K 18.9%, 100K-250K 25.5%, 250K+ 40.9% ($p < 0.001$). The top 1% of transactions (average NGN 518,390) are flagged 35.8% of the time against 19.4% for the rest ($p < 0.001$).
Business Meaning	Current review practice already responds to transaction size. The 250K+ tier is only 4.2% of transactions, so it is a small, high-exposure group that is cheap to monitor closely.
Recommended Action	Keep value-based review thresholds, and report the flag rate by value tier on the dashboard so any drift in review coverage at the high end is visible.

Insight 3: About 8% of transactions fail or reverse, and no channel is clearly worse	
Finding	7.97% of transactions end as Failed (5.25%) or Reversed (2.72%), and the rate is similar across channels.
Evidence	956 failed or reversed transactions, worth NGN 40.45M (7.2% of total value). By channel: ATM 6.9%, Web 7.1%, POS 7.7%, Mobile App 8.6%, USSD 8.7%. The channel difference is not significant at the 5% level ($p = 0.07$). The highest channel-and-type cells (e.g. USSD Cash Withdrawal 10.0%, 100 transactions) are not significantly above the 8% average ($p = 0.46$). Success rate overall is 90.47%.
Business Meaning	Failures look like a platform-wide reliability level rather than a problem in one channel. Roughly 1 in 12 transactions does not complete cleanly, which affects customer trust and support workload.
Recommended Action	Investigate failure and reversal causes across the whole platform (the dataset has no failure-reason field, so add one), and monitor channel rates monthly to catch any channel that moves clearly above the average.

Insight 4: Customer segments differ in size, not in behaviour	
Finding	Segments do not separate on risk or on spend per customer. Everyday leads totals because it has the most customers.
Evidence	Everyday holds 47.4% of customers and 46.6% of transaction value. Spend per customer: Everyday NGN 367.7K, Premium 365.3K, Student 386.6K, SME 387.9K. Transactions per customer 7.9 to 8.2 in every segment. Risk-review rate 19.3% to 20.3% across segments ($p = 0.75$).
Business Meaning	Segment labels do not predict how customers transact, so segment-based targeting or risk rules would treat very similar customers differently. Student and SME customers are as valuable per head as Everyday.
Recommended Action	Build behaviour-based groups (transaction frequency, total spend, preferred channel) for marketing and retention, and use the segment label only for reporting size and mix.

Insight 5: Mobile App carries the most traffic and Transfer carries the most value	
Finding	Mobile App handles 42.5% of transactions, and Transfer generates 42.4% of all value.
Evidence	Channel share of transactions: Mobile App 42.5%, POS 19.9%, Web 15.6%, ATM 14.6%, USSD 7.4%. Transfer is NGN 237.9M of NGN 560.5M total (29.6% of transactions) and is the top transaction type by value in every segment.
Business Meaning	The digital core of FinTrust is the Mobile App Transfer journey. Problems there affect the largest share of customers and value, while USSD is a small channel.
Recommended Action	Prioritise Mobile App and Transfer in reliability work and customer support capacity, and keep USSD as a lower-priority, access-focused channel.

Insight 6: The February dip is a calendar effect, not a slowdown	
Finding	Monthly value fell in February only because the month is shorter. Daily activity is flat to slightly rising.
Evidence	Monthly value: Jan NGN 188.5M, Feb 175.8M, Mar 196.2M. Per day: Jan 6.08M, Feb 6.28M, Mar 6.33M (about +4% Jan to Mar). Transactions per day are identical at 133 in all three months. Day-of-week volume is even (1,602 to 1,734 per day).
Business Meaning	Comparing raw monthly totals would suggest a decline that is not real. There is also no weekly pattern to plan staffing around.
Recommended Action	Use average daily value as the trend KPI in Power BI alongside monthly totals, and avoid claiming growth or decline from three months of data. Collect a longer history before trend conclusions.

Findings that changed during final review

Re-testing earlier conclusions with significance tests changed three of them. These changes feed directly into Part E (Final Validation).

Earlier claim (Weeks 2-3)	What the final check showed	Revised conclusion
ATM + International (46.0%) is the highest-risk combination and should be auto-flagged	Only 63 transactions; interval 34% to 58%; channel differences within international are not significant (p = 0.27)	Risk is driven by international status as a whole (36.9% vs 18.9%). ATM is directional only.
Failures and reversals are concentrated in USSD	USSD 8.7% vs 7.9% for other channels; cell-level 10% figures are within normal variation (p = 0.42 to 0.46)	Failure is platform-wide at about 8%. No channel is clearly worse.
February dip may be a genuine fluctuation	Transactions per day are identical in all months; February has 28 days	The dip is a calendar effect. Daily value is slightly rising.

Limitations. Three months of data (Jan to Mar 2026), so seasonality and long-term trends cannot be assessed. International transactions are 480 records, so any breakdown within them is small-sample. The data is synthetic. Risk_Review_Flag shows which transactions were sent for review, not confirmed fraud, so these insights describe review activity rather than actual loss.